Chapter test

38 Microbiology

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. The protein coat of a virus encloses

   A. DNA only.
   B. RNA only.
   C. DNA or RNA.
   D. DNA and RNA. ☐

2. The following events occur when a virus infects a host cell.

   (1) synthesis of protein coat
   (2) lysis of host cell
   (3) penetration into host cell
   (4) synthesis of viral genetic material
   (5) taking over the metabolism of host cell

   Which of the following shows the correct sequence of events?

   A. (3) → (5) → (4) → (1) → (2)
   B. (3) → (4) → (5) → (1) → (2)
   C. (3) → (5) → (1) → (2) → (4)
   D. (4) → (1) → (5) → (3) → (2) ☐

★ 3. Which of the following statements about bacteria is ***incorrect***?

   A. All bacteria are prokaryotes.
   B. Bacteria may have more than one plasmid.
   C. The chromosomes of bacteria are located inside the nucleus.
   D. Bacteria can be autotrophic. ☐

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4. Which of the following factors affect the growth of microorganisms?

(1) availability of carbon
(2) availability of nitrogen
(3) availability of water

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

5. A student decided to carry out an experiment to find out if a species of bacterium produces protease or not. Which of the following is a suitable culturing method?

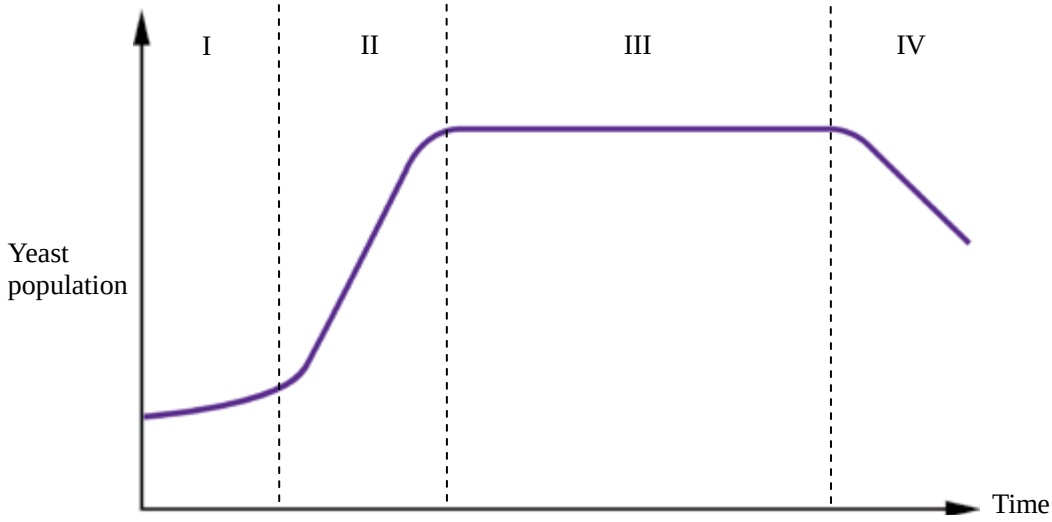
A. incubating the bacteria in a nutrient broth with milk
B. incubating the bacteria in a nutrient broth with essential amino acids
C. incubating the bacteria on an agar plate with milk
D. incubating the bacteria on an agar plate with an antibiotic ☐

★ 6. Which of the following is ***not*** one of the safety precautions that must be taken when handling microorganisms in laboratory?

A. Open all the windows and doors of the laboratory during experiment.
B. Disinfect the working area after the experiment.
C. Wear a laboratory coat and disposable gloves.
D. Use sterilized and autoclaved apparatus. ☐

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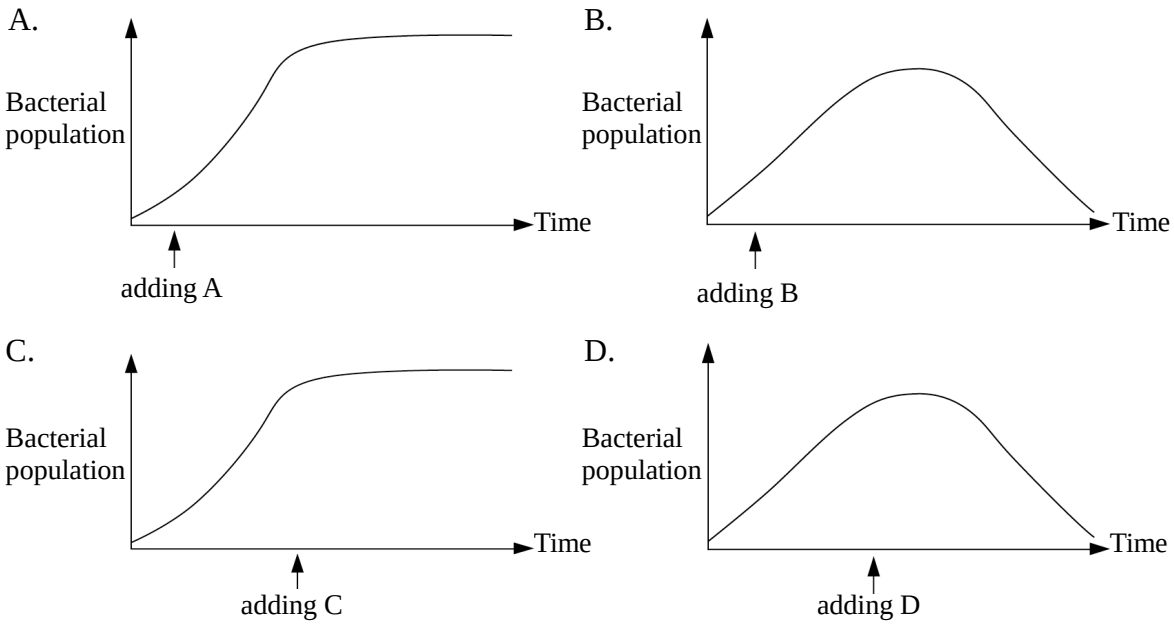
★ 7. The graph below shows the growth curve of a yeast population in a nutrient broth.



During which phase do some of the nutrients in the broth start to run out?

A. I
B. II
C. III
D. IV

8. In an investigation, a scientist tested the effect of four antibiotics, A, B, C and D, towards a type of bacterium by adding them into a bacterial culture respectively. The graphs below show the results. Which antibiotic is the most effective towards this type of bacterium?



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★ 9. Which of the following correctly compares the various measurement methods of microbial growth?

(1) Cell counts measure the number of microorganisms per unit area while biomass method measures the mass of microorganisms per unit volume.
(2) Both total cell count and optical method count living and dead microorganisms.
(3) Biomass method involves separating microorganisms from the liquid culture while optical method does not.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

★★10. An experiment began with two bacterial cells and ended up with 512 cells. How many generations did the bacterial species go through in the experiment?

A. 256
B. 160
C. 9
D. 8 ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	38.2	38.2	38.3	38.4	38.5	38.5	38.6	38.6	38.6	38.6

B. Problem solving (23 marks)

11. A student introduced a pure culture of bacteria into a nutrient broth. He collected a sample from the broth and measured the number of bacteria after 24 hours. Serial dilution is carried out before counting the number of bacteria.

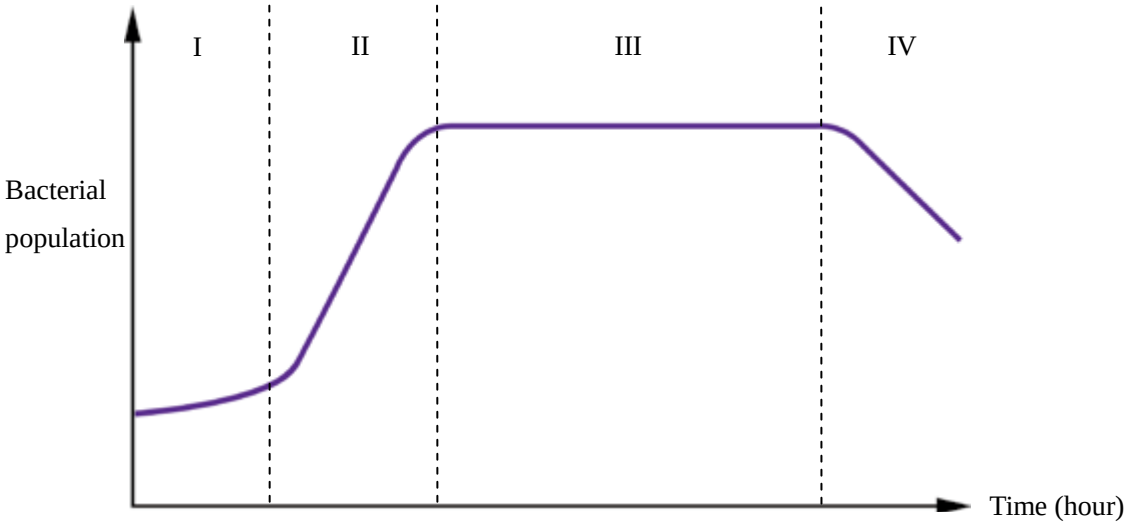
(a) (i) Suggest why it is necessary to carry out serial dilution. (1 mark)

(ii) Describe the procedures of serial dilution. (3 marks)

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(iii) The concentration of bacteria in the sample after five times of 10-fold dilution is 75 bacterial cells per cm^3. If the original culture is 30 cm^3, calculate the number of bacteria in the original culture. Show your working. (2 marks)

(b) The graph below shows the growth curve of the bacteria.



(i) Name phases II and IV. (2 marks)

(ii) Describe and explain the shape of the curve in phases I and III. (4 marks)

(Total: 12 marks)

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★ 12. Microorganisms in a liquid culture can be inoculated into a solid culture medium using streak plate method.

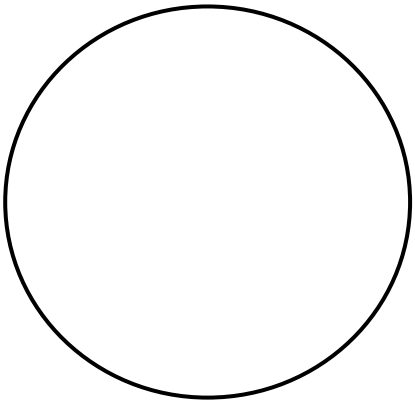
(a) What is the name of the above process? (1 mark)

(b) (i) Give ***one*** example of solid culture medium commonly used to grow microorganisms. (1 mark)

(ii) Describe ***one*** advantage of using a solid culture medium instead of a liquid culture medium. (1 mark)

(c) Briefly describe the steps involved in obtaining microorganisms from the liquid culture. (5 marks)

(d) After obtaining microorganisms from the liquid culture, the microorganisms are spread onto the solid culture medium by streaking the plate containing the medium three times. The plate is then incubated for one day. Sketch the pattern of the streaks in the circle below. (3 marks)



(Total: 11 marks)

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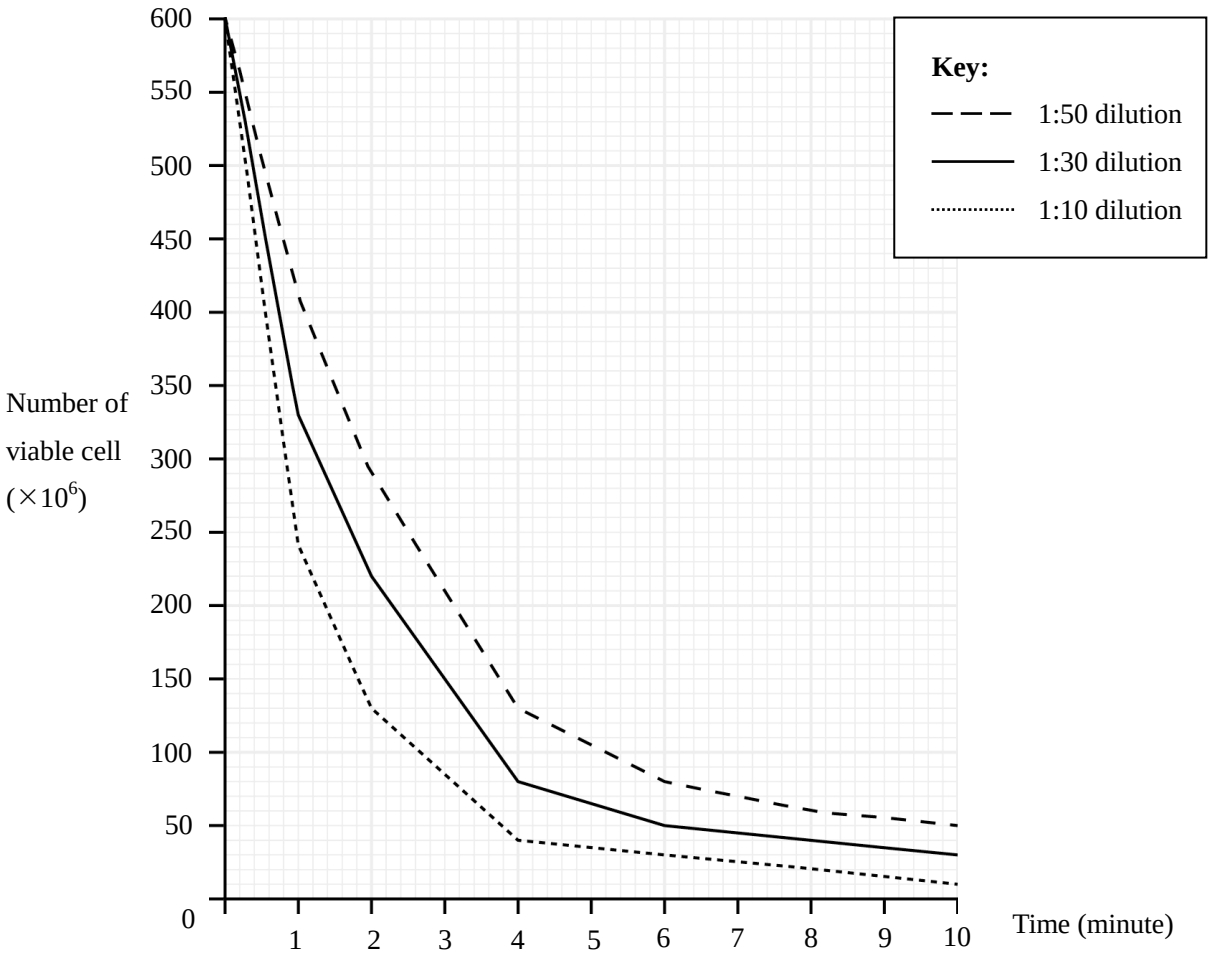
C. Exam challenge (7 marks)

★ 13. In microbiology laboratory, all apparatus and equipment have to be sterilized before use. People working in the laboratory should also follow a set of safety procedures.

(a) What is the meaning of sterilization? (1 mark)

(b) Before starting an experiment, microbiologists have to carry out risk assessment on the procedures of the experiment. Suggest ***two*** aspects that should be taken into consideration in risk assessment to lower the risk associated with the handling of bacteria. (2 marks)

(c) A microbiologist accidentally spilt a culture containing pathogenic bacteria. He used a disinfectant to clean up the spillage. Disinfectants are often diluted before use. The graph below shows the number of viable bacterial cells left after different dilutions of the disinfectant have been applied.



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(i) The spillage is considered to be cleared up when the number of viable cells is less than 80×10^6. The microbiologist left the disinfectant on the spillage for 5 minutes. Suggest which dilution is the most appropriate to use. Explain your answer.

(3 marks)

(ii) Suggest a disinfectant that can be used to clean up the spillage. (1 mark)

(Total: 7 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	C	2.	A	3.	C	4.	D	5.	C
6.	A	7.	C	8.	D	9.	C	10.	D

(1 mark each)

B. Problem solving (23 marks)

11. (a) (i) The number of cells in the original culture is too large, making it difficult to count when grown on an agar plate. (1)

(ii) A known volume (e.g. 1 cm^3) is taken from the culture. Dilute it by adding the sample into 9 cm^3 of sterile culture medium. Mix well. (1)

The diluted sample is then taken (e.g. 1 cm^3) and diluted in the same way. (1)

The process is repeated to produce a series of solutions of decreasing concentration. (1)

(iii) Number of bacterial cell = $75 \times 10^5 \times 30$ (1)

$= 2.25 \times 10^8$ (1)

(b) (i) II: exponential phase (1)

IV: death phase (1)

(ii) In phase I, as the bacterial cells are adapting to the new environment, the growth rate is low. (1)

Therefore, the slope of the curve is small. (1)

In phase III, some nutrients begin to run out and metabolic wastes accumulate. This makes population growth slow down. The rate of cell formation is equal to the rate of death. (1)

Therefore, the slope of the curve at this phase appears to be zero. (1)

(Total: 12 marks)

12. (a) Inoculation (1)

(b) (i) Nutrient agar plate (1)

(Or other acceptable answers)

(ii) It will not be broken down by the microorganisms growing on it. / As agar is translucent, it provides a background against which colonies of microorganisms can be seen easily. (1)

(c) Dip an inoculating loop into 70% ethanol. Heat the inoculating loop over a Bunsen flame until it is red hot. (1)

Allow the loop to cool for 10 seconds. (1)

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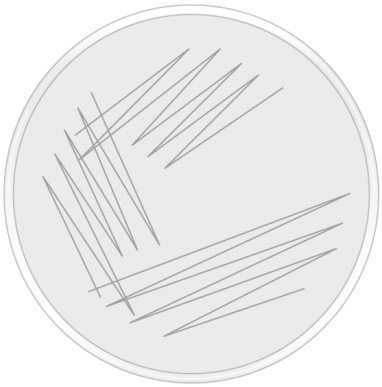
Open the bottle containing the liquid culture and flame the mouth of the bottle over the Bunsen flame. (1)

Dip the inoculating loop into the liquid culture to withdraw a small amount of the sample. (1)

Flame the mouth of bottle again and close the bottle. (1)

(d) Correct drawing: (3)

- Three sets of streaks (1)
- The next set of streak is perpendicular to the previous set (1)
- The next set of streak came from the previous set (with an overlap region) (1)



(Total: 11 marks)

C. Exam challenge (7 marks)

13. (a) Kill all the microorganisms and their spores. (1)

(b) Any ***two*** of the following: (2)

- Choice of microorganism, e.g. use of non-pathogenic bacteria (1)
- Choice of growth medium, e.g. with antibiotics to kill undesirable microorganisms (1)
- Exposure time to the microorganisms, e.g. seal the incubated culture in a transparent plastic bag when observing the results (1)

(Or other acceptable answers)

(c) (i) 1 : 30 dilution (1)

Both 1:10 and 1:30 dilutions can reduce the number of viable cells to less than 80×10^6 within 5 minutes. (1)

The 1:30 dilution has a lower concentration of disinfectant and is more environmentally friendly / less disinfectant is used to save cost / is safer to use. (1)

(ii) 70% ethanol / Chlorine bleach (1)

(Total: 7 marks)

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